Department of Computer Applications

II MCA – Java Programming Lab

Java Lab Practice – Jagged Arrays & ArrayList

## Problem 1: Classroom Marks Analyzer

**Digital Story:**

A teacher records marks of students from different sections. Each section has a different number of students. Use a jagged array to store marks and compute section-wise averages.

**Input:**

Number of sections, number of students in each section, marks of each student.

**Sample I/O:**

Input:
3
2 3 1
78 90
45 56 67
88

Output:
Section 1 Avg = 84
Section 2 Avg = 56
Section 3 Avg = 88

**Test Cases:**

1) 1 section, 1 student     average = mark
2) Sections with varying sizes
3) Zero students in a section should be skipped

**Skeleton Code:**

```
public class MarksAnalyzer {
  public static void main(String[] args) {
    // read input
    // create int[][] jagged array
    // calculate averages
  }
}
```

**Viva Questions:**

1) What is a jagged array?
2) Why not use 2D fixed array here?
3) Time complexity of average calculation?

## Problem 2: Library Shelf Management

**Digital Story:**

A library has shelves with different numbers of books. Store titles using ArrayList of ArrayList and display all books shelf-wise.

**Sample I/O:**

Input:
2 shelves
Shelf1: Java Python
Shelf2: DBMS OS CN

Output:
Shelf1 -> [Java, Python]
Shelf2 -> [DBMS, OS, CN]

**Test Cases:**

1) Empty shelf
2) Single book shelf
3) Many shelves

**Skeleton Code:**

```
import java.util.*;

public class Library {
  public static void main(String[] args) {
    ArrayList<ArrayList<String>> shelves = new ArrayList<>();
    // add books
  }
}
```

**Viva Questions:**

1) Difference between ArrayList and array?
2) How dynamic resizing works?
3) How to iterate nested lists?

## Problem 3: Cricket Tournament Scoreboard

**Digital Story:**

Each team plays a different number of matches. Store runs using jagged arrays and display highest scorer per team.

**Sample I/O:**

Input:
Team1: 50 40 30
Team2: 80 10

Output:
Team1 max = 50
Team2 max = 80

**Test Cases:**

1) Single match
2) All equal scores
3) Negative/invalid score handling

**Skeleton Code:**

```
public class Tournament {
  public static void main(String[] args) {
    int[][] scores;
    // find max per row
  }
}
```

**Viva Questions:**

1) How memory is allocated for jagged array?
2) How to find max efficiently?
3) What happens if row is null?

## Problem 4: Shopping Cart System

**Digital Story:**

Customers buy different numbers of products. Use ArrayList to store prices and compute total bill for each customer.

**Sample I/O:**

Input:

Customer1: 100 200
Customer2: 50 20 30

Output:
Total1 = 300
Total2 = 100

**Test Cases:**

1) Empty cart
2) Single item
3) Large number of items

**Skeleton Code:**

```
import java.util.*;

public class ShoppingCart {
  public static void main(String[] args) {
    ArrayList<ArrayList<Integer>> carts = new ArrayList<>();
    // calculate totals
  }
}
```

**Viva Questions:**

1) Why use ArrayList over array here?
2) How to sum using loop?
3) Can we use streams?

## Problem 5: College Timetable Planner

**Digital Story:**

Each day has a different number of classes. Use jagged array to store subjects and print timetable neatly.

**Sample I/O:**

Input:
Mon: 3 classes
Tue: 2 classes

Output:
Mon -> Math Physics Java
Tue -> DBMS OS

**Test Cases:**

1) No class day
2) Maximum classes
3) Different subject counts

**Skeleton Code:**

```
public class Timetable {
  public static void main(String[] args) {
    String[][] schedule;
    // print timetable
  }
}
```

**Viva Questions:**

1) Difference between jagged and rectangular arrays?
2) How to traverse 2D array?
3) Can we replace with ArrayList? Why?